ML4DD Final Project - Lipophilicity

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Abstract

Lipophilicity is a main component in drug design, in fact as Schroeter et al. 2007 state in the very first words of their article underlining the subject, "Unfavourable lipophilicity and water solubility cause many drug failures".

In this project we aim to study machine learning modelling techniques to give us an accurate understanding of what compounds will be adequately soluble and will therefore fare well as drug compounds, reproducing techniques used by experts in the field. We describe how we studied 3 non-GNN regression models and 3 GNN regression models, with R^2 scores ranging from 0.303 to 0.607 in the non-GNN models, and reaching up to 0.823 in the best GNN application, and RMSEs ranging from 2.243 to 0.520 in the non-GNN models, and reaching a best value of 0.5 with the best GNN model; and how this compares to previous, expert, approaches.

1 Introduction

The ability of a drug to interact with the human body can be measured by scoring several properties, including the absorption, distribution, metabolism, and excretion (ADME) properties. ADME is an acronym used mainly in fields such as pharmacokinetics and pharmacology.

The first property, absorption, is a measure that identifies the bioavailability of a compound, measured through impacting factors including lipophilicity, the ability of a compound to dissolve in a lipid environment. This factor determines cell permeability and the compound's ability to be absorbed into the bloodstream, therefore the ability of a drug to travel from the site of administration to the site of action.

Lipophilicity is measured using a partition coefficient, P, which refers to the ratio of a dissolved compound in a two-phase system; and a distribution coefficient, D, which represents the partition coefficient of the compound in both neutral and ionized form at any pH (Bhal 2024; Najafi and Cheu 2017). Both these measures are usually given in the form of logarithm base 10, as "LogP" and "LogD", with the most widespread and relevant descriptor being "Log D7.4", the distribution coefficient of a compound at a pH level of 7.4 (neutral).

In general, higher lipophilicity will lead to better membrane permeability but poorer aqueous solubility, and vice versa.

2 Materials

2.1 Data

The Lipophilicity AstraZeneca dataset was obtained via the Therapeutics Data Commons (TDC) Python API (Huang et al., 2021). It contains 4,200 drug-like molecules with experimentally measured logD values at pH 7.4, originally curated by AstraZeneca (Wenlock and Tomkinson 2016) and distributed as part of the MoleculeNet benchmark suite (Wu et al. 2017).

The target variable is the octanol-water distribution coefficient logD, which accounts for the ionization state of molecules at physiological pH, making it more biologically relevant than the simple partition coefficient logP (Bhal 2024).

2.2 Data Preprocessing

SMILES strings were converted to RDKit molecule objects, canonicalized and deduplicated. No molecules were removed from the AstraZeneca dataset. No logD range filter was applied (Wenlock and Tomkinson 2016). Missing values in the physicochemical descriptor matrix were imputed using column medians computed on the training set only.

2.3 Molecular Representations

Four encodings were computed from canonical SMILES using RDKit: MACCS Keys (166 bits), Morgan Fingerprints/ECFP4 (1024 bits, radius=2) (Rogers and Hahn 2010), RDKit Fingerprints (2048 bits, maxPath=7), and an extended set of 101 physicochemical descriptors covering lipophilicity, size, hydrogen bonding, polarity, topology, ring systems, atom counts, electronic properties and functional groups. Physicochemical descriptors were standardized using z-score normalization fitted on the training set only.

2.4 Data Splitting and Similarity Analysis

The dataset was split 70/10/20 into training, validation and test sets using TDC's standardized splits (Huang et al. 2021) under two strategies: random

split (baseline, comparable to MoleculeNet) and scaffold split (primary evaluation, molecules grouped by Murcko scaffold). Nearest-neighbour Tanimoto similarity of each test molecule to the training set was computed using Morgan fingerprints to quantify split difficulty and support applicability domain analysis.

3 Methodology

3.1 Feature Selection

Clustered Forward Selection (CFS) was applied to the physicochemical descriptors using a Spearman correlation threshold of 0.7 to group features into clusters, then performing forward selection with at most one candidate per cluster. The optimal feature count was determined by the peak cross-validated R^2 score. CFS was applied separately for each model on the training set only.

3.2 Classic Models

Three classical models were evaluated: Linear Regression (baseline), Random Forest, and XGBoost, trained across all four encodings and both splits (24 combinations). All experiments used a fixed random seed (42) for reproducibility. As a bonus extension, Kernel Ridge Regression was evaluated with two variants — Tanimoto kernel on Morgan fingerprints and Gaussian kernel on CFS-selected physicochemical descriptors — motivated by prior lipophilicity prediction work (Schroeter et al. 2007). Graph Neural Networks (GCN and GIN) were also evaluated as a bonus extension and are described in Section 3.3.

3.3 Graph Models

Graph Neural Networks (GNNs) model each molecule as a graph G=(V,E), where atoms are nodes and chemical bonds are edges. At each layer, node representations are updated by aggregating information from neighboring atoms, allowing the model to learn local chemical environments and construct a global molecular embedding.

Two primary architectures were evaluated.

Graph Convolutional Networks (GCN): Utilising the update rule:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l}W^{(l)}) \quad (1)$$

where $\tilde{A} = A + I$ adds self-loops to the adjacency matrix and $\tilde{D}$ is its degree matrix. The normalisation stabilises training and encourages smooth passing across bonds.

Graph Isomorphism Networks(GIN): Utilising a learnable MLP as the update rule:

$$h_v^{(k)} = MLP^{(k)} * \left((1+\epsilon)h_v^{(k)} + \sum_{u \in N(v)} h_u^{(k-1)} \right) \quad (2)$$

After several message-passing layers, node embeddings are summed (instead of pooled) into a fixed-length graph embedding and passed to an MLP head for regression.

We compared three specific implementations to contrast "from-scratch" development with optimised libraries (both on random split and scaffold split as for previous models):

- GCN (Scratch): A custom dense implementation using one-hot atomic number features (54 dims) and a batch size of 1.
- PyG GCN: Implemented via PyTorch Geometric using OGB's smiles2graph featurisation and sparse batching (size 64).
- PyG GIN: Utilising GINConv with the same high-dimensional OGB atom and bond features.

3.4 Hyperparameter Tuning

Linear Regression requires no tuning. Random Forest and XGBoost were tuned via grid search on the validation set minimising MAE. The search grids were as follows:

- Random Forest: n_estimator (50, 100, 200, 500), max_depth (None, 5, 10, 20), and min_samples_leaf (1, 2, 5) — 48 combinations total.
- XGBoost: n_estimators (50, 100, 200, 500), max_depth (3, 5, 7, 9), and learning_rate (0.01, 0.05, 0.1, 0.15) — 64 combinations total.

KRR lambda was searched over 6 values for Tanimoto and 16 combinations for Gaussian.

3.5 Evaluation Metrics

Models were evaluated using MAE (primary), RMSE (comparable to MoleculeNet benchmark, Wu et al. 2017) and R^2.

3.6 Application Domain

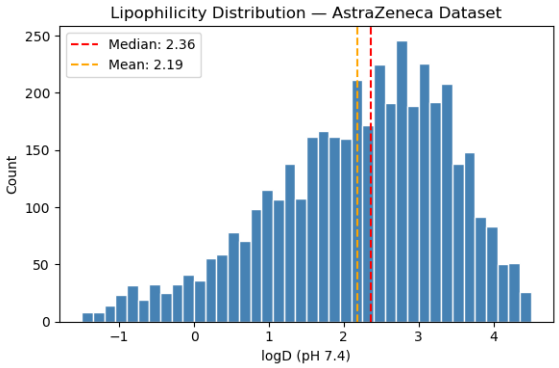
Test molecules were grouped into four Tanimoto similarity bins (¡0.4, 0.4–0.5, 0.5–0.7, ¿0.7) and performance was computed per bin for the best model. The AD threshold was determined empirically as the similarity value below which error increased substantially. Bootstrap analysis (n=1,000 resamples) was performed to verify the threshold was not driven by sample size differences.

4 Results

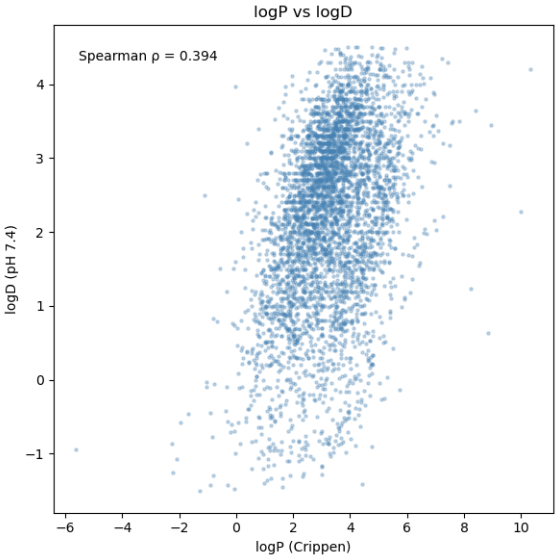
4.1 Dataset Characterisation, Exploratory Analysis and Chemical Space

The logD target distribution is approximately bell-shaped and slightly left-skewed (mean=2.19, median=2.36, std=1.20, range 1.5 to 4.5), with the majority of compounds concentrated in the moderately

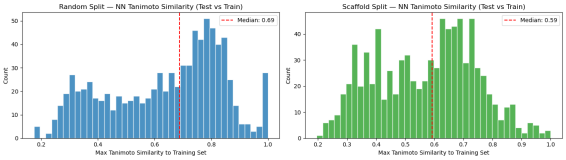
lipophilic range of logD 1–3.5. Median molecular weight of 388.4 Da, logP of 3.3, H-bond donors of 1 and acceptors of 5 confirm the dataset represents typical drug-like chemical space (Lipinski et al. 2001).



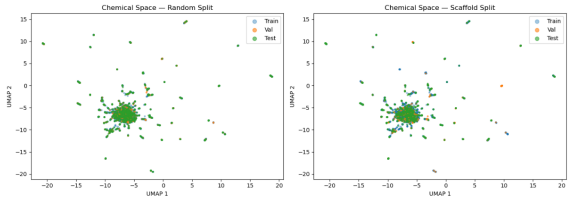
Despite their close relation, the Spearman correlation between logP and logD is only moderate (=0.394), and the wide vertical spread at any given logP value confirms that logP alone is not sufficient for accurate logD prediction, motivating the use of additional descriptors¨.



The 4,200 molecules were split into training (2,940), validation (420) and test (840) sets. Nearest-neighbour Tanimoto similarity analysis reveals a mean of 0.639 and median of 0.688 for the random split, compared to 0.571 and 0.593 for the scaffold split, confirming that scaffold splitting creates a slightly harder evaluation. The difference is moderate (0.1 drop in median), suggesting relatively homogeneous chemical space.

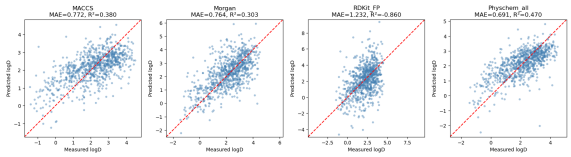


UMAP visualisation confirms this — in the random split, test molecules dominate both core and peripheral regions, while in the scaffold split all three partitions are relatively more evenly distributed across chemical space, consistent with the lower median similarity.

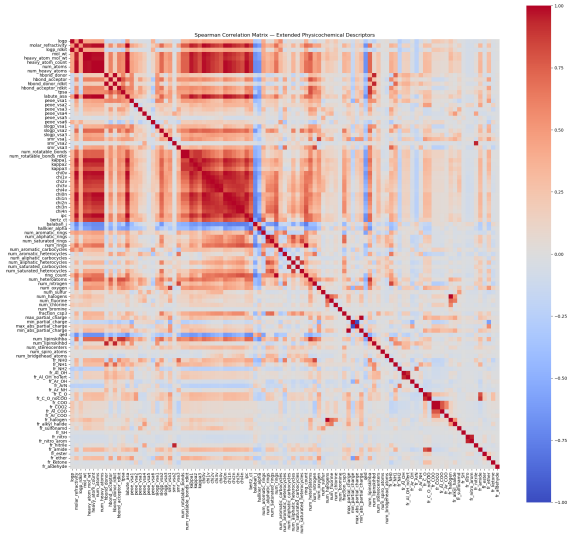


4.2 Baseline Model and Feature Selection

As a diagnostic step before the full model comparison, we trained Linear Regression on all features from each encoding on the random split. The scatter plots reveal clear visual differences: MACCS and Morgan show reasonable but noisy predictions clustering around the diagonal, physicochemical descriptors show the tightest scatter, while RDKit fingerprints produce a completely failed model with predictions compressed into a narrow horizontal band far from the diagonal.



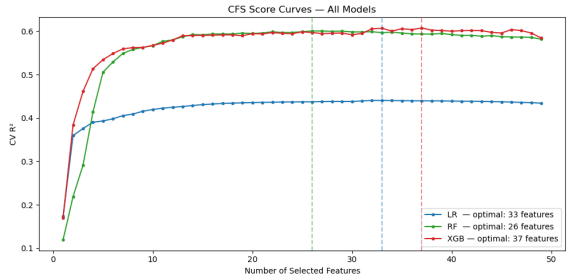
Numerically, physicochemical descriptors performed best (MAE=0.691, R^2=0.470), with MACCS and Morgan showing moderate performance (R^2=0.380 and 0.303 respectively). While RDKit fingerprints produced a severely negative R^2 = 0.860 due to their extremely high feature density of 846 active bits per molecule (41.3% of 2048 bits), creating an ill-conditioned problem for linear regression, Morgan and MACCS fingerprints had far lower feature densities of 48 (4.7%) and 52 (31.1%) active bits respectively, making them more tractable for linear models.



For the physicochemical descriptors, inspection of the Spearman correlation matrix revealed strong redundancy in the 101-descriptor with correlated clusters among connectivity/topological indices, molecular size descriptors, and lipophilicity variants, motivating CFS. Applied separately per model, CFS identified 49 correlated clusters and selected 33, 26, and 37 features for Linear Regression, Random Forest, and XGBoost re-

spectively, achieving cross-validated R^2 of 0.440, 0.601, and 0.607.

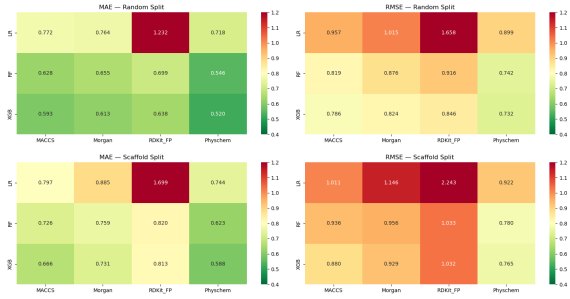
The selected feature sets differ meaningfully across models. $\log p_r dkit appear sin all three, confirming it as the mos$ $tion, alongside$ $related descriptors, surface area terms, and hydrogen bond do$ $a2', 'fr_A r_O H 'a$ $linear signals that only tree - based models can exploit.$



Performance between the full and CFS-selected descriptor sets was negligible for Linear Regression (R^2 within 0.025), confirming that CFS removed redundancy without sacrificing predictive power. PFI analysis further validated this — $\log p_r dkit and fr_C OO emerged as the dominant descriptors post$ $CFS, resolving the importance dilution observed across the fu$

4.3 Full Model Comparison

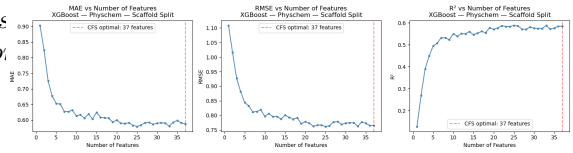
We trained and evaluated 24 model/encoding/split combinations (3 models × 4 encodings × 2 splits), summarised via MAE and RMSE heatmaps below.



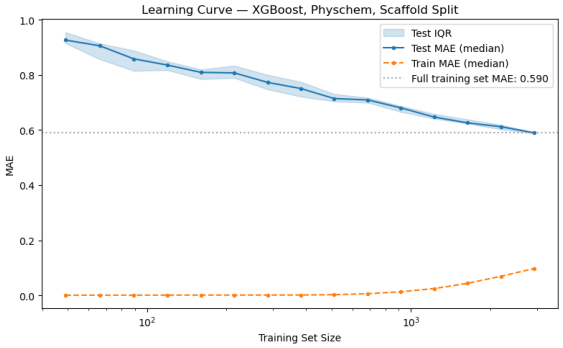
The results reveal consistent patterns. Model ranking is consistent across all encodings and splits: XGBoost ¿ Random Forest ¿ Linear Regression. Encoding ranking is consistent across most combinations: Physchem ¿ MACCS ¿ Morgan ¿ RDKit$_FP, said otherwise, the CFS -$ $selected physicochemical descriptors outperform all fingerprint encodings, suggesting that explicit chemical knowledge is more$

The best overall model is XGBoost with physicochemical descriptors, achieving MAE=0.520, RMSE=0.732, R^2=0.637 (random split) and MAE=0.588, RMSE=0.765, R^2=0.584 (scaffold split). The scaffold split consistently yields higher MAE across all models, confirming that generalising to unseen chemical series is harder. Linear Regression suffers most (MAE increase of 0.025–0.467) while XGBoost is most robust (0.068–0.175), suggesting tree-based models generalise better to new scaffolds.

We further investigated feature count sensitivity for the best: performance improves steeply for the first 5–10 features then plateaus, with an optimal count of 25 (MAE=0.580) (marginally fewer than the 37 selected by CFS). The negligible difference (change in MAE=0.008) confirms the full CFS-selected set is justified, with most predictive signal captured by the first 20–25 features.



The learning curve shows consistent improvement from MAE=0.927 at n=49 to MAE=0.590 at n=2,939 — an improvement of 0.337 MAE units. The curve has not plateaued at the full training set size, suggesting the dataset is in a data-limited regime and additional logD measurements would meaningfully improve performance.

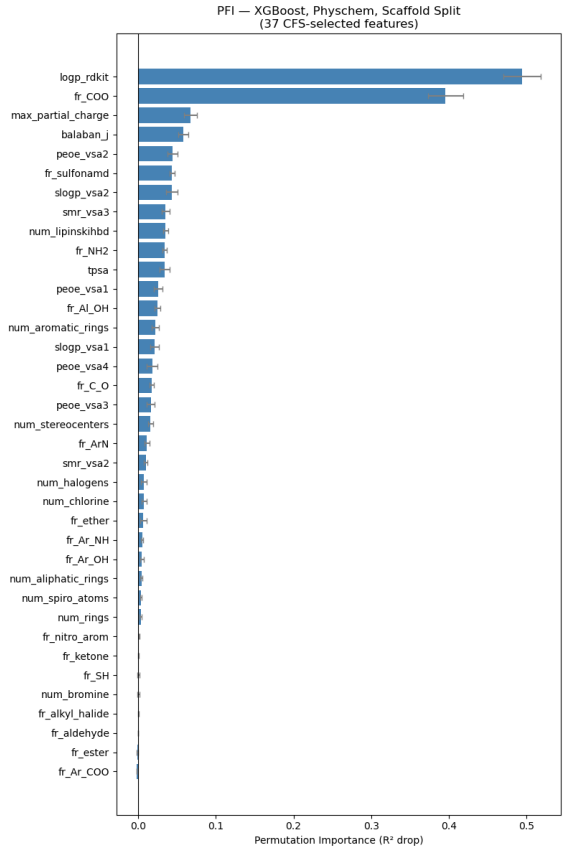


4.4 Performance vs Similarity, Applicability Domain and Explainability

To investigate whether prediction error increases with structural distance from the training set, scaffold test molecules were grouped into four Tanimoto similarity bins. MAE and RMSE increase monotonically as similarity decreases — from MAE=0.470, RMSE=0.587 $(\text{nn}_s im > 0.7, n = 207) to MAE = 0.707, RMSE = 0.921 (nn_s im < 0.4, n = 174). A slight R non - monotonicity between the two lowest bins (0.565 vs 0.535) was attributed$ $1,000 resamples), with substantially overlapping 95\% confidence inte$

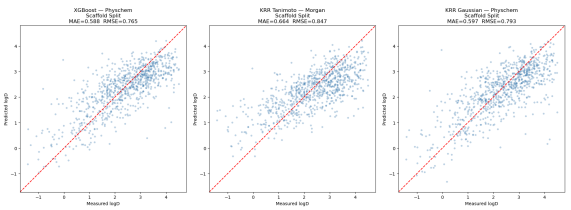
Based on this analysis, we set the applicability domain threshold at $\text{nn}_s im = 0.5$, $where the largest single MAE drop occurs (0.660ß0.572). This place$ $0.533 and RMSE = 0.692, versus MAE = 0.687 and RMSE = 0.881 outside, a 29\% MAE increase, confirmings$

PFI on the best model reveals a clear hierarchy: $\log p_r dkit(0.494) and fr_C OO(0.396) dominate, jointly accounting for t$ $zero, confirming that CFS ensured independent contributions. The lo$ $importance tail likely captures structural edge cases rather than gener$



4.5 Kernel Ridge Regression

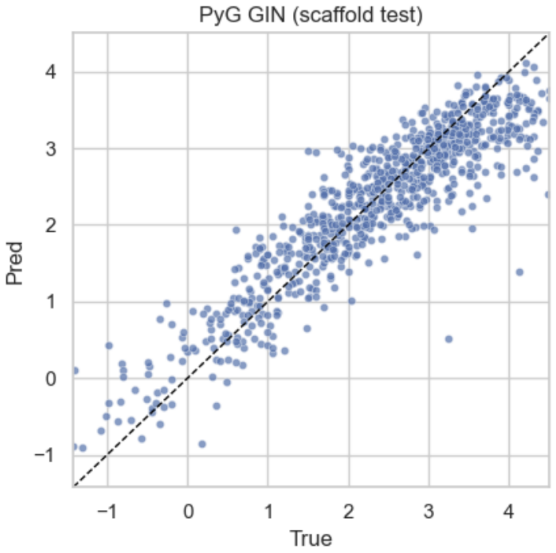
As a bonus extension, we evaluated two KRR variants on the scaffold split. KRR Tanimoto with Morgan fingerprints achieved MAE=0.664 and RMSE=0.847 (=0.1), while KRR Gaussian with the same CFS-selected physicochemical descriptors as XGBoost achieved MAE=0.597 and RMSE=0.793 (=0.1, lengthscale=5.0). The Gaussian variant comes remarkably close to XGBoost (MAE=0.588) — a difference of only 0.009 MAE units, despite being a much simpler model with a single lengthscale parameter. KRR Tanimoto shows noticeably more scatter in the predictions, consistent with Morgan fingerprints being less informative for logD than physicochemical descriptors.



4.6 Graph Model Results

GNN performance was evaluated across both random and scaffold splits. While the classical XGBoost models performed well, the GNNs, specifically the GIN architecture, demonstrated the highest predictive power on the scaffold split.

Split	Model	RMSE	MAE	R2
Random	GCN (scratch)	0.878	0.691	0.478
Random	PyG GCN	0.713	0.554	0.656
Random	PyG GIN	0.667	0.508	0.699
Scaffold	GCN (scratch)	0.904	0.714	0.420
Scaffold	PyG GCN	0.666	0.524	0.686
Scaffold	**PyG GIN**	**0.500**	**0.379**	**0.823**



The scratch GCN was the weakest model because of a few clear technical reasons.

First, it doesn't have much information to work with; it only looks at the atomic number, missing out on important chemical details like aromaticity and hybridisation. The PyG models, on the other hand, use a much richer set of features.

Second, the scratch model oversimplifies chemical bonds into a basic "yes or no" connection, while the PyG models look at the specific types of bonds.

The way the models gather information from surrounding atoms also matters. The standard GCN tends to "blur" the details of a molecule by using means, while the GIN model is much better at keeping track of unique local structures, like specific functional groups, by summing and using non-linear MLP.

When looking at the results, the PyG GCN and GIN models did better on the harder scaffold split than on the random one. Even though this is unusual compared to standard models, it shows that GNNs are very good at recognising structural patterns when they are given high-quality data.

5 Discussion

In this work, we investigated the prediction of LogD7.4 values for 4,200 compounds using three classical machine learning models and three graph neural network (GNN) architectures. Our best non-GNN model (XGBoost) achieved an RMSE of 0.732, while the best GNN model (GIN) reached an RMSE of 0.50 on the AstraZeneca dataset. Compared to prior studies, such as Wu et al. 2017 (RMSE 0.65) and Lenselink and Stouten 2021 (RMSE = 0.66 for general-purpose models and 0.39 for challenge-specific models), our GNN

performance appears competitive, while classical models remain within expected ranges.

These results should be interpreted cautiously. The strong performance, particularly of the GNN, is likely influenced by the relatively limited dataset size and composition. Model performance in lipophilicity prediction is highly dependent on data quantity and diversity, and a larger, more chemically varied dataset would likely provide a more realistic evaluation of generalizability.

An unexpected observation was that the best GNN performance was obtained using a scaffold-based split rather than a random split, which is typically considered more favorable. This may indicate that the model learned more generalizable structural features, although further validation is required.

Our approach follows common practices in the literature, where models such as Random Forest are used as strong nonlinear baselines. Previous studies, including Schroeter et al. 2007, highlight the value of comparing diverse models such as Ridge Regression, Support Vector Regression, and Gaussian Processes, the latter offering advantages in uncertainty estimation. Similarly, Lenselink and Stouten 2021 demonstrate the effectiveness of advanced strategies such as ensembling and using model predictions as input features, achieving state-of-the-art performance.

Overall, model success in this domain depends on several key factors: dataset size and quality, data splitting strategy, molecular representation, and model selection. Among these, data remains the most critical component.

In conclusion, both classical models and GNNs are effective for lipophilicity prediction, with GNNs showing particular promise in capturing structural information. While our results are competitive, future work should focus on expanding datasets, improving evaluation strategies, and exploring hybrid or ensemble approaches to achieve more robust and generalizable predictions.

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